

Problem1

1. Sampling CT sinusoids

Consider 3 CT signals:

$$\begin{aligned} x_1(t) &= \cos(3000t), \\ x_2(t) &= \cos(4000t), \text{ and} \\ x_3(t) &= \cos(5000t). \end{aligned}$$

Each of these is sampled as follows

$$\begin{aligned} x_1[n] &= x_1(nT), \\ x_2[n] &= x_2(nT), \text{ and} \\ x_3[n] &= x_3(nT), \end{aligned}$$

where $T = 0.001$. Which of the resulting DT signals has the highest DT frequency? Which has the lowest DT frequency?

Problem 1. $x_1[n] = \cos(3n)$ $\omega T = 2\pi, \therefore \omega \in (-\pi, \pi)$ $\left\{ \begin{array}{l} \omega = 0/2\pi \dots \text{最低频率}(1) \\ \omega = \pi/3\pi \dots \text{最高频率}(\pm 1) \end{array} \right.$
 $x_2[n] = \cos(4n) = \cos[(4-2\pi)n] = \cos[(2\pi-4)n] = \cos(2.28n)$
 $x_3[n] = \cos(5n) = \cos[(2\pi-5)n] = \cos(1.28n)$
 $\Rightarrow$ 具有最高DT频率的是 $x_1[n]$, 最低DT频率: $x_3[n]$. (越靠近 $\omega=0$, 频率越低)
 (假如取 $\omega \in (0, 2\pi)$, 越靠近 $\omega=\pi$, f 越高)

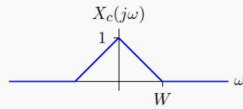
Problem2

2. Sampling with alternating impulses

A CT signal $x_c(t)$ is converted to a DT signal $x_d[n]$ as follows:

$$x_d[n] = \begin{cases} x_c(nT) & n \text{ even} \\ -x_c(nT) & n \text{ odd} \end{cases}$$

a. Assume that the Fourier transform of $x_c(t)$ is $X_c(j\omega)$ shown below.



Determine the DT Fourier transform $X_d(e^{j\Omega})$ of $x_d[n]$.

b. Assume that $x_c(t)$ is bandlimited to $-W \leq \omega \leq W$. Determine the maximum value of W for which the original signal $x_c(t)$ can be reconstructed from the samples $x_d[n]$.

c. Make a diagram of a system to reconstruct $x_c(t)$ from $x_d[n]$.

Problem 2 已知 $x_d[n] = \begin{cases} x_c(nT), & n \text{ 偶} \\ -x_c(nT), & n \text{ 奇} \end{cases}$ $x_c: \text{CT}, x_d: \text{DT}$.

(a) 已知 $X_c(j\omega)$: 求 $X_d(e^{j\Omega})$

$$\text{令 } x_e[n] = x_c(nT), \text{ 则 } X_e(e^{j\Omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j\frac{\Omega - 2k\pi}{T})$$

$$x_d[n] = x_e[n] \cdot (-1)^n = x_e[n] \cdot e^{j\pi n} \Rightarrow X_d(e^{j\Omega}) = X_e(e^{j(\Omega - \pi)}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j\frac{\Omega - (2k+1)\pi}{T})$$

(b) 假设 $x_c(t)$ 是带限信号, $\omega \in [-W, W]$.

求原始信号 $x_c(t)$ 能从样本 $x_d[n]$ 中完全重建的 W 最大值.

$$\frac{2\pi}{T} = \omega_s > 2W \Rightarrow W_{\max} = \frac{\pi}{T}.$$

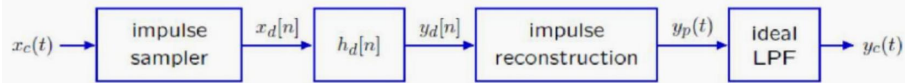
(c) 画出从 $x_d[n]$ 重建 $x_c(t)$ 的系统框图.

$$x_d[n] \xrightarrow{x(-1)^n} x_c(nT) \rightarrow x_c(t), \text{ 其中 } x_c(t) = \frac{1}{\pi} \sum_{n=-\infty}^{\infty} x_c(nT) \frac{\sin[\omega_s(t-nT)]}{t-nT}.$$

Problem3

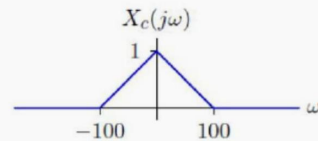
DT processing of CT signals

Sampling and reconstruction allow us to process CT signals using digital electronics as shown in the following figure.



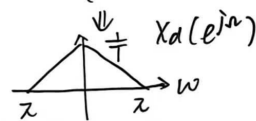
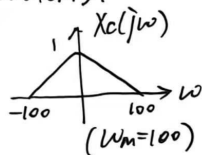
The "impulse sampler" and "impulse reconstruction" use sampling interval $T = \pi/100$. The unit-sample function $h_d[n]$ represents the unit-sample response of an ideal DT low-pass filter with gain of 1 for frequencies in the range $-\frac{\pi}{2} < \Omega < \frac{\pi}{2}$. The "ideal LPF" passes frequencies in the range $-100 < \omega < 100$. It also has a gain of T throughout its pass band.

Assume that the Fourier transform of the input $x_c(t)$ is $X(j\omega)$ shown below.



Determine $Y_c(j\omega)$.

Problem3.



$X_c(t) \rightarrow$ 冲激采样器 $\rightarrow X_d[n] \rightarrow h_d[n] \rightarrow y_d[n] \rightarrow$ 冲激重建 $\rightarrow y_p(t) \rightarrow$ 低通滤波器 $\rightarrow y_c(t)$
 $T = \frac{\pi}{100}$ 求 $Y_c(j\omega)$ n 在 $(-\frac{\pi}{2}, \frac{\pi}{2})$ 为 1.
 采样: $\omega_s = \frac{2\pi}{T} = 200 \text{ rad/s} \gg 2\omega_m$ 无混叠.

已知 $X_d(e^{jn})$, $H_d(e^{jn})$, $Y_d(e^{jn}) = X_d(e^{jn}) \cdot H_d(e^{jn})$

$$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n] \delta(t - nT)$$

$$\Rightarrow Y_p(j\omega) = Y_d(e^{j\omega T})$$

$$Y_c(j\omega) = Y_p(j\omega) \cdot H(j\omega) = \begin{cases} T \cdot Y_d(e^{j\omega T}), & |\omega| < 100 \\ 0, & |\omega| > 100 \end{cases}$$

$$\omega_0 = \frac{2\pi}{T} = 50 \text{ rad/s},$$

$$Y_c(j\omega) = \begin{cases} T \cdot \frac{1}{T} X_c(j\omega) = X_c(j\omega), & |\omega| \leq 50 \\ 0, & |\omega| > 50 \end{cases}$$



Problem4

7.6. In the system shown in Figure P7.6, two functions of time, $x_1(t)$ and $x_2(t)$, are multiplied together, and the product $w(t)$ is sampled by a periodic impulse train. $x_1(t)$ is band limited to ω_1 , and $x_2(t)$ is band limited to ω_2 ; that is,

$$X_1(j\omega) = 0, |\omega| \geq \omega_1,$$

$$X_2(j\omega) = 0, |\omega| \geq \omega_2.$$

Determine the *maximum* sampling interval T such that $w(t)$ is recoverable from $w_p(t)$ through the use of an ideal lowpass filter.

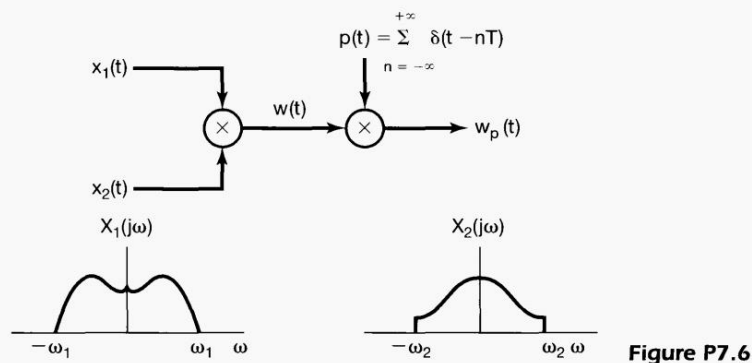


Figure P7.6

Problem 4

$$w(t) = x_1(t) \cdot x_2(t)$$

$$\Rightarrow W(j\omega) = \frac{1}{2\pi} X_1(j\omega) * X_2(j\omega) \Rightarrow \text{最高频率 } \omega_m = \omega_1 + \omega_2.$$

$$\frac{2\pi}{T} \geq 2(\omega_1 + \omega_2) \Rightarrow T \leq \frac{\pi}{\omega_1 + \omega_2}$$

Problem5

7.23. Shown in Figure P7.23 is a system in which the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is as indicated in the figure.

- For $\Delta < \pi/(2\omega_M)$, sketch the Fourier transform of $x_p(t)$ and $y(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $x_p(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $y(t)$.
- What is the maximum value of Δ in relation to ω_M for which $x(t)$ can be recovered from either $x_p(t)$ or $y(t)$?

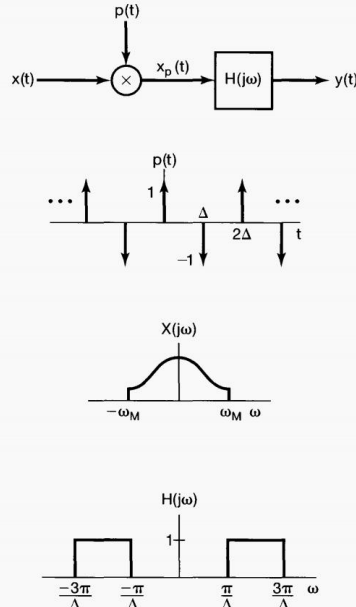
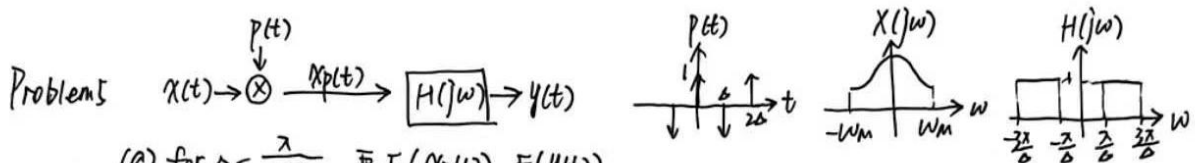


Figure P7.23

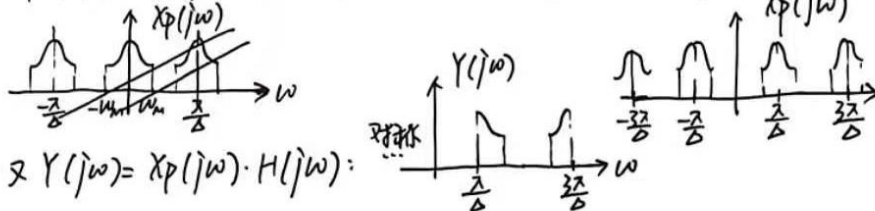


(a) for $\Delta < \frac{\pi}{2\omega_M}$, $\mathcal{F}\{x_p(t)\}$, $\mathcal{F}\{y(t)\}$

$$p(t) = \sum_{n=-\infty}^{\infty} (-1)^n \delta(t - n\Delta) \Rightarrow \frac{2\pi}{\Delta} \sum_{k=-\infty}^{\infty} \delta(\omega - (2k+1)\frac{\pi}{\Delta}) = P(j\omega)$$

$$X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega) = \frac{1}{\Delta} \sum_{k=-\infty}^{\infty} X[j(\omega - (2k+1)\frac{\pi}{\Delta})]$$

平移中后对称轴: $\pm \frac{\pi}{\Delta}, \pm \frac{3\pi}{\Delta}, \dots$ 又 $\omega_M < \frac{\pi}{2\Delta} \Rightarrow$ 互不影响



$$\text{又 } Y(j\omega) = X_p(j\omega) \cdot H(j\omega):$$

(b) 确定一个能从 $x_p(t)$ 中恢复 $x(t)$ 的系统.

$$X_p(j\omega) \text{ 在 } \omega=0 \text{ 处 } = 0. \quad x_p(t) \xrightarrow{\times \cos(\frac{\pi}{2\Delta}t)} \text{通过} \begin{matrix} \uparrow \\ \text{低通滤波器} \end{matrix} \rightarrow x(t)$$

(c) 从 $y(t)$ 恢复 $x(t)$ 的系统.

$$y(t) \xrightarrow{\times p(t)} \text{通过} \begin{matrix} \uparrow \\ \text{低通滤波器} \end{matrix} \rightarrow x(t)$$

(d) $\Delta_{\max}$.

$$\frac{2\pi}{\Delta} > 2\omega_M \Rightarrow \Delta_{\max} = \frac{\pi}{\omega_M}$$